
PROJECT BRIEF

AI Underwriting Analyst

Build a production-style AI agent in three weeks

FOR

Lead developers

DURATION

3 weeks

DOMAIN

Fintech · loan underwriting

MODE

Sandbox · no real actions

SECTION 01

What you are building

You will build an AI agent that reviews a loan application and recommends what to do with it.

It reads the applicant's documents — bank statements, income details, and KYC — finds the important numbers, checks them against a lending policy, looks for warning signs, and gives a clear recommendation: **approve, refer, or decline**.

It also produces two files: a short underwriting memo (PDF) and a cash-flow summary (Excel).

A person reviews the result. **The agent only reads and recommends**. It never approves, sends, or changes anything.

SECTION 02

Why this project

Your last project was about finding information (RAG). This one is about doing work.

You will build an agent that plans its own steps, splits a big job into smaller ones, keeps its memory under control during a long task, and checks that its answers are right. These are the everyday skills for building AI products that run in production.

SECTION 03

What you will practice

The project is built so that you have to use each idea below. You are done with an idea when you can show the result in the last column.

Idea	What it means here	You are done when
Agents for long tasks	The agent makes its own to-do list, hands small jobs to helper agents, and keeps going across many steps.	It plans its own steps, finishes a long run, and picks up where it left off if it is stopped halfway.
Skills	The lending policy lives in its own file the agent reads when needed, instead of being written into prompts. The agent also uses ready-made skills to create the Excel and PDF.	Changing a rule means editing one file, not the prompts. The Excel and PDF are real, usable files.
Checking the output	You build a set of test applications with known correct answers, and a script that scores the agent against them.	One command runs the tests and prints a score report.
Using fewer tokens	You stop sending the same large text again and again: save documents to files, reuse cached parts, and do the number-crunching in code.	Each application uses far fewer tokens than before, with no drop in the score.
Picking the right model	Use a small, cheap model for simple jobs (reading fields) and a strong model only for the hard thinking (the decision and the memo).	Each step runs on the cheapest model that still passes the tests, and you can explain why.
Keeping memory small	Long bank statements are summarised between steps and kept in notes, so the main agent's memory stays small.	The agent's memory stays small even when the documents get longer.

One smaller task: try a few document readers (parsers) on the same statements and choose the best one for each document type. Treat how well they read as something you measure. Spend a few days on this, not a week.

Underneath all of it: record every step so you can see what happened, handle bad or missing data without crashing, and keep an eye on cost.

SECTION 04

How it works — the steps

The agent moves through five steps. A main agent plans the run and hands jobs to helper agents.

- Read the documents** — turn each file into clean text; pick the best reader for each file type.
- Pull out the numbers** — a helper agent per document finds income, transactions, and ID details. (small model)

- Check and calculate** — compare the documents, then calculate the ratios in code. (small model)
- Make the decision** — apply the lending policy, weigh the warning signs, decide approve / refer / decline. (strong model)
- Write the outputs** — the memo (PDF), the cash-flow summary (Excel), and the decision. (skills)

SECTION 05

What you are given

- The lending policy** — the rules the agent must follow. Turn this into the policy Skill.
- Sample applicant packets** — made-up applications to build and test against. Each packet has a bank statement (PDF), a KYC and credit file (PDF), and an income sheet (Excel).

Nothing in the data is real. There is no real personal information.

SECTION 06

What to deliver

At the end of week 3, hand in **two things**:

1. A working demo — run the agent on a fresh application it has not seen and show it produce the decision, the memo, and the Excel. It should recover cleanly if it is stopped midway.

2. A short report — covering:

- the test scores: how often the decision was right, how accurate the numbers were, the memo quality, tokens per application, cost per application, and speed;
- the before-and-after on tokens and cost from your clean-up pass;
- which model you used for each step, and why;
- which document reader you chose for each file type, and why.

You are done when the demo runs and the report is handed in.

SECTION 07

A suggested schedule

When	Focus	Goal for the week
Week 1	Get the basics working: build test data with known answers; try the document readers; build the agent skeleton that plans and pulls out fields; turn on tracing.	One application goes through, start to finish.

When	Focus	Goal for the week
Week 2	Build the real agent: full planning and helper agents; the policy Skill; the Excel and PDF outputs; summarising long statements; small + strong model split; tests on every run.	A correct memo on applications it has not seen.
Week 3	Prove it and cut cost: make the tests a gate; reduce tokens and cost; handle bad input and mid-run stops; finish the report; demo.	Demo runs and report is handed in.

If you finish early: let the agent learn from a reviewer's corrections; add a human approval step before the memo; add tricky, fraud-like applications to your tests.

SECTION 08

Rules to follow

- Only read and recommend.** Never approve, send, or change anything.
- Use the sample data only.** Do not put real personal data into the system.
- Do not get stuck** on document readers or setup. The agent and the testing are what matter.